

D910

Data Analysis for Healthcare Decisions

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

An analytics team at a health system that just moved to a cloud data warehouse reviews length of stay for 500 pneumonia patients. Most stayed 3 to 5 days, but a few stayed more than 40 days. The report goes to the board chair, a retired banker. Which measure best represents a typical stay?

- A. The mean, because a banker expects averages
- B. The mean
- C. The range
- D. The median

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A survey at a clinic with a newly renovated waiting area asks patients to rate their pain as none, mild, moderate, or severe. The survey was translated into four languages. What level of measurement is this?

- A. Nominal
- B. Interval, because the categories look evenly spaced
- C. Ordinal
- D. Ratio

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Unit A, a 30-bed unit with a new nurse manager, had 12 patient falls in a quarter with 2,400 patient days. Unit B, which was recently repainted, had 9 falls with 1,500 patient days. Which unit has the higher fall rate per 1,000 patient days?

- A. They are equal at 5.0 per 1,000
- B. Unit A, because it has a new manager and more beds
- C. Unit A, because 12 falls is more than 9
- D. Unit B, 6.0 vs. 5.0 per 1,000 patient days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A unit's goal is a fall rate of no more than 3.0 per 1,000 patient days. It expects 4,000 patient days next quarter. What is the most falls it can have and still meet the goal?

- A. 3
- B. 12
- C. 120
- D. 1,333

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A hospital's 30-day readmission rate fell from 20% to 15% after a new discharge program led by a pharmacist with a PhD. The hospital also updated its logo that year. How should the change be described?

- A. A 25-percentage-point decrease
- B. A 33% decrease
- C. A 5% decrease
- D. A 5-percentage-point drop, which is a 25% relative decrease

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Clinic visits rose 40% over last year to 2,100. How many visits were there last year?

- A. 840
- B. 1,260
- C. 1,500
- D. 2,940

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A study at a system known for its sports medicine program tested an automated appointment reminder with 80,000 patients. No-show rates fell from 10.0% to 9.8%, with $p = 0.01$. The vendor offers a free trial. What is the best interpretation?

- A. There is a 99% chance the reminder works
- B. Adopt it, because the trial is free
- C. The drop is statistically significant but very small; weigh its practical value against cost
- D. The effect is large and important because p is below 0.05

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A pilot with 40 patients found a 6-point drop in no-shows, with $p = 0.20$. What is the best interpretation?

- A.** The result is not statistically significant and could be due to chance; a larger study is needed
- B.** It is significant because a 6-point drop is large
- C.** 20% of patients benefited from the reminder
- D.** The reminder has been proven to have no effect

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An analyst at a health system with a new chief data officer builds a model that estimates each patient's probability of readmission within 30 days, using past records. The chief data officer holds an MBA. Which type of analytics is this?

- A.** Predictive
- B.** Diagnostic
- C.** Prescriptive
- D.** Descriptive, since any analysis of past data is descriptive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A quality director at a hospital that recently joined a larger system wants to show monthly ED wait times for the past 24 months to see whether they are trending up. The system's colors are blue and gold. Which chart fits best?

- A.** A histogram of all wait times
- B.** A line (run) chart of monthly wait times
- C.** A pie chart of wait times by month
- D.** A scatter plot of wait time against staffing

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A sepsis alert tested at a hospital with a new rapid-response team flagged 90 of 100 patients who truly had sepsis. It also flagged 180 of 900 patients who did not. The vendor is based overseas. What is the alert's sensitivity?

- A. 20%
- B. 90%
- C. 80%
- D. 33%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

Using the same results (90 of 100 true cases flagged, 180 of 900 non-cases flagged), what share of flagged patients truly had sepsis?

- A. 33%
- B. 80%
- C. 10%
- D. 90%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

An analytics team at a health system ~~that just moved to a cloud data warehouse~~ reviews length of stay for 500 pneumonia patients. Most stayed 3 to 5 days, but a few stayed more than 40 days. The report goes to the board chair, ~~a retired banker~~. Which measure best represents a typical stay?

Target word: best center measure **Decides:** Right-skewed data with outliers.

A	Camouflage	The audience does not change which measure fits skewed data.
B	Confusion set	The mean is pulled upward by the long outliers.
C	Adjacent category	The range describes spread, not the typical value.
D	Correct	The median resists the pull of the few very long stays.

2. Answer C Concept 2

A survey at a clinic ~~with a newly renovated waiting area~~ asks patients to rate their pain as none, mild, moderate, or severe. ~~The survey was translated into four languages.~~ What level of measurement is this?

Target word: measurement level **Decides:** Ranked categories without known equal spacing.

A	Confusion set	Nominal categories have no order; these do.
B	Dropped element	Drops the fact that equal spacing between labels is not established.
C	Correct	The categories have an order but unknown spacing.
D	Adjacent category	Ratio data are measured amounts with a true zero.

3. Answer D Concept 3

Unit A, ~~a 30-bed unit with a new nurse manager~~, had 12 patient falls in a quarter with 2,400 patient days. Unit B, ~~which was recently repainted~~, had 9 falls with 1,500 patient days. Which unit has the higher fall rate per 1,000 patient days?

Target word: higher fall rate **Decides:** Rates: $12/2,400$ vs. $9/1,500$.

A	Wrong stem number	Uses Unit A's patient days for both units.
B	Camouflage	Management and bed count do not determine the rate.
C	Dropped element	Drops the patient-day denominators.
D	Correct	A: $12 \div 2,400 \times 1,000 = 5.0$; B: $9 \div 1,500 \times 1,000 = 6.0$.

4. Answer B Concept 3 • transfer

A unit's goal is a fall rate of no more than 3.0 per 1,000 patient days. It expects 4,000 patient days next quarter. What is the most falls it can have and still meet the goal?

Target word: maximum falls **Decides:** $3.0 \times (4,000 \div 1,000)$.

A	Dropped element	Drops the scaling from 1,000 to 4,000 patient days.
B	Correct	$3.0 \times 4 = 12$ falls.
C	Wrong stem number	Scales as if the rate were per 100 patient days.
D	Inverted operation	Divides patient days by the rate instead of multiplying.

5. Answer D Concept 4

A hospital's 30-day readmission rate fell from 20% to 15% after a new discharge program led by a pharmacist with a PhD. The hospital also updated its logo that year. How should the change be described?

Target word: how to describe the change **Decides:** Point change vs. relative change.

A	Adjacent category	Labels the relative change as a point change.
B	Inverted operation	Divides by the new value (15) instead of the old (20).
C	Confusion set	Confuses percentage points with percent change.
D	Correct	$20 - 15 = 5$ points; $5 \div 20 = 25\%$.

6. Answer C Concept 4 • transfer

Clinic visits rose 40% over last year to 2,100. How many visits were there last year?

Target word: prior-year value **Decides:** $2,100 \div 1.40$.

A	Confusion set	Treats 40% of the new total as the old total.
B	Inverted operation	Subtracts 40% of 2,100 instead of dividing by 1.4.
C	Correct	$2,100 \div 1.4 = 1,500$.
D	Wrong stem number	Applies the increase again to the new number.

7. Answer C Concept 5

A study at a system known for its sports medicine program tested an automated appointment reminder with 80,000 patients. No-show rates fell from 10.0% to 9.8%, with $p = 0.01$. The vendor offers a free trial. What is the best interpretation?

Target word: interpretation **Decides:** Significant but very small effect.

A	Confusion set	A p-value is not the probability the program works.
B	Camouflage	The free trial does not answer whether the effect matters.
C	Correct	A large sample can make a 0.2-point change significant without making it important.
D	Dropped element	Drops the effect size, which is only 0.2 points.

8. Answer A Concept 5 • transfer

A pilot with 40 patients found a 6-point drop in no-shows, with $p = 0.20$. What is the best interpretation?

Target word: interpretation **Decides:** Not significant; small sample.

A	Correct	A large p-value means the data can't rule out chance, not that there is no effect.
B	Confusion set	Confuses practical size with statistical significance.
C	Adjacent category	A p-value is not a share of patients.
D	Absolute language	A non-significant result does not prove zero effect.

9. Answer A Concept 6

An analyst at a health system ~~with a new chief data officer~~ builds a model that estimates each patient's probability of readmission within 30 days, using past records. ~~The chief data officer holds an MBA.~~ Which type of analytics is this?

Target word: analytics type **Decides:** Estimates the likelihood of a future event.

A	Correct	Estimating what is likely to happen is predictive.
B	Confusion set	Diagnostic explains why something already happened.
C	Adjacent category	Prescriptive recommends actions; this model only scores risk.
D	Absolute language	Using past data to forecast is predictive.

10. Answer B Concept 7

A quality director at a hospital ~~that recently joined a larger system~~ wants to show monthly ED wait times for the past 24 months to see whether they are trending up. ~~The system's colors are blue and gold.~~ Which chart fits best?

Target word: chart choice **Decides:** A trend over time.

A	Confusion set	A histogram shows the distribution but loses time order.
B	Correct	Line charts show change across time.
C	Adjacent category	Pie charts show parts of a whole, not trends.
D	Dropped element	Answers a different question and drops the time order.

11. Answer B Concept 8

A sepsis alert tested at a hospital ~~with a new rapid response team~~ flagged 90 of 100 patients who truly had sepsis. It also flagged 180 of 900 patients who did not. ~~The vendor is based overseas.~~ What is the alert's sensitivity?

Target word: sensitivity **Decides:** 90 of 100 true cases flagged.

A	Inverted operation	That is the false-positive rate: $180 \div 900$.
B	Correct	Sensitivity = $90 \div 100 = 90\%$.
C	Confusion set	That is specificity: $720 \div 900$.
D	Adjacent category	That is the positive predictive value: $90 \div 270$.

12. Answer A Concept 8 · transfer

Using the same results (90 of 100 true cases flagged, 180 of 900 non-cases flagged), what share of flagged patients truly had sepsis?

Target word: PPV **Decides:** $90 \div (90 + 180)$.

A	Correct	$PPV = 90 \div 270 \approx 33\%$.
B	Adjacent category	That is specificity.
C	Wrong stem number	Divides 90 by the 900 non-cases.
D	Confusion set	That is sensitivity.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Mean vs. median in skewed data	When data are skewed by a few extreme values, the median represents the typical case better than the mean.	Reporting the mean length of stay when a few outliers pull it up.	"Most were 3-5 days, but a few stayed 40+."
2	Levels of measurement	Nominal names categories, ordinal ranks them without equal spacing, interval has equal spacing without a true zero, and ratio has a true zero.	Treating ranked categories like mild-moderate-severe as evenly spaced numbers.	Ordered labels versus counts or measured amounts.
3	Rates need denominators	Compare units with rates (events ÷ exposure × a base such as 1,000 patient days), not raw counts.	Calling the unit with more events the worse unit.	Two units with different volumes.
4	Percentage points vs. percent change	A move from 20% to 15% is a 5-percentage-point drop and a 25% relative decrease; percent change = (new – old) ÷ old.	Calling a 5-point drop a 5% decrease, or dividing by the new value.	Two percentages compared over time.
5	Statistical vs. practical significance	A small p-value means the result is unlikely to be chance alone; it does not mean the effect is large or worth acting on, and a large p-value does not prove no effect.	Reading p = 0.01 as a 99% chance the program works, or p > 0.05 as proof of no effect.	A p-value paired with a tiny or large effect size.
6	Types of analytics	Descriptive tells what happened, diagnostic why, predictive what is likely, and prescriptive what to do.	Calling any model built on past data descriptive.	"Estimate the probability" versus "recommend the action."
7	Matching the chart to the question	Use a line or run chart for trends over time, a bar chart to compare categories, a histogram for a distribution, and a scatter plot for a relationship between two measures.	Using a pie or histogram when the question is about change over time.	"Over the past 24 months" or "by department."
8	Sensitivity, specificity, and PPV	Sensitivity = true positives ÷ all with the condition; specificity = true negatives ÷ all without it; PPV = true positives ÷ all who tested positive.	Swapping sensitivity with PPV or specificity.	A screening tool's counts of flagged and unflagged patients.

Course scope (public description): Data Analysis for Healthcare Decisions is designed to introduce learners to the fundamental principles and techniques of data analysis within the context of healthcare decision-making. This course aims to prepare learners to analyze healthcare data, interpret findings, and apply these insights to improve patient outcomes, optimize healthcare operations, and inform evidence-based decision-making. There are no prerequisites to the course.